

Chapter II

A. Questions and answers

1. Explain the difference between the process and the program.

A:

The process is the active entity, and the program is the passive entity.

Processes are temporary, and programs are persistent.

The process is different from the content of the program.

2. Please describe the main functions and differences of the various schedulers.

A:

Long-range scheduler: Responsible for selecting the appropriate program from the disk and putting it into memory ready for execution.

Mid-range scheduler: Responsible for temporarily moving some processes out of memory to secondary memory to free up some memory space, and then move back in after a period of time to wait for execution.

Short-range scheduler : Responsible for allocating the CPU to processes that have entered memory and are ready to be executed.

3. Please explain what status is usually available for itineraries?

A:

Created: The state when the process was first created.

Ready: The itinerary is ready and ready to be executed.

Suspended: The journey is waiting for an event to occur, usually waiting for the I/O operation to complete.

Execution: The process is running, that is, the CPU is being used.

Terminated: The trip has ended, but some information remains in the system.

Displace/Ready: Procedures can be executed, but are currently overridden in secondary memory.

Displace/Suspend: The process is waiting for an event to occur and is overridden into secondary memory.

4. Describe the circumstances under which a new itinerary is generated.

A:

User login

The user runs a program

Request the system to provide services

generated by an existing itinerary

5. Please explain the possible reasons for the end of the tour.

A:

Task completion, protection error, calculation error, I/O failure, external force termination, parent process termination, parent process request.

6. Describe what information about the trip is typically stored in the Trip Control section.

A:

Schedule status and schedule information

Memory management

Hardware status

Input/output status information

Additional control information: trip ID, trip ID of the parent trip, access permissions, start time, etc.

7. Briefly describe common queues in the operating system and explain their purpose.

A:

Job queue: When a process is allowed to enter the system, it is placed in the job queue.

Ready Queue: When a process is ready to run, it is moved to the ready queue.

Wait Queue: Processes that wait for certain events are moved into a wait queue.

Device Queue: Processes waiting for I/O devices are moved to the device queue.

8. Please explain what is body switching?

A:

When CPU usage is switched from one process to another, the operating system must store information about the process in the program control section of that process and load the program control section of the other process into the system in order to restore the execution environment to the state it was in when the latter was interrupted. This action is called body switching.

12. When a thread is created, what resources are used? How is it different from creating an itinerary?

A:

Threads must have their own program counters, registers, and stack space. But unlike processes, all threads in the same process share the same memory address space, program segments, data sections, and some other system resources, such as open files and signals.

B. Calculation problems

1. There is a set of itineraries as follows:

Process CPU burst time (ms)

P1	15
P2	7
P3	4
P4	10

The arrival time is 0 and is in the order of P1, P2, P3 and P4.

Use FCFS and non-preemptive SJF scheduling methods to find the average wait time, respectively.

A:

Average Wait Time for FCFS:

$$(0 + 15 + 22 + 26) / 4 = 15.75 \text{ ms}$$

Average wait time for SJF:

$$(0 + 4 + 11 + 21) / 4 = 9 \text{ milliseconds}$$

2. Suppose there are four processes P1, P2, P3, and P4, all of which arrive at time 0, in the order of P1, P2, P3, and P4, and their CPU burst times are as follows:

Process CPU burst time (ms) priority

P1	6	2
P2	5	0
P3	7	3

P4

4

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- a. When using FCFS, SJF, non-preemptive priority (the lower the number, the higher the priority), and RR (time slice = 3) scheduling methods, please draw a diagram similar to the text about the order in which the procedures are executed.
- b. What is the average wait time for all trips?
- c. What is the response time for each trip?

A:

a.

The schedule for FCFS is as follows:

P1 P2 P3 P4

0 6 11 18 22

The SJF schedule is as follows:

P4 P2 P1 P3

0 4 9 15 22

The schedule for non-preemptive priority is as follows:

P2 P4 P1 P3

0 5 9 15 22

The schedule for RR is as follows:

P1 P2 P3 P4 P1 P2 P3 P4 P3

0 3 6 9 12 15 17 20 21 22

b.

Average Wait Time for FCFS:

$$(0 + 6 + 11 + 18) / 4 = 8.75 \text{ ms}$$

Average wait time for SJF:

$$(0 + 4 + 9 + 15) / 4 = 7 \text{ ms}$$

Average wait time for non-preemptive priority:

$$(0 + 5 + 9 + 15) / 4 = 7.25 \text{ ms}$$

Average waiting time for RR:

$$((12-3) + (15-3) + (20 - 3) + (21 - 6)) / 4 = 13.25 \text{ ms}$$

c.

	FCFS	SJF	Priority	RR
P1	6	15	15	15
P2	11	9	5	17
P3	18	22	22	22
P4	22	4	9	21

3. Suppose there are four processes P1, P2, P3, P4, and P5, all of which arrive at time 0, in the order of P1, P2, P3, P4, and P5, and their CPU burst times are as follows:

Process CPU burst time (ms) priority

P1	10	3
P2	3	1

	P3	2	3
P4	1	4	
	P5	5	2

(a) When using FCFS, SJF, non-preemptive priority (the lower the number, the higher the priority), and RR (time slice = 3, executed in order of arrival) scheduling methods, please draw a diagram similar to the text about the order in which the itinerary is executed.

(b) What is the average wait time for each trip in the various algorithms mentioned above? Which algorithm has the lowest average wait time?

(c) If you want RR to perform better, according to the rule of thumb, what should be the minimum time slicing in the execution case of the above example?

A:

(b) FCFS=10.8, SJF=4.2, priority=9.8, RR=8; The average time of SJF is the shortest

(c) 5 milliseconds